



Perfect out-forest problem and directed Steiner cycle packing problem

Yuefang Sun^a, Zemin Jin^{b,*}

^a School of Mathematics and Statistics, Ningbo University, Ningbo 315211, PR China

^b School of Mathematical Sciences, Zhejiang Normal University, Jinhua 321004, PR China

ARTICLE INFO

Article history:

Received 10 January 2024

Received in revised form 17 December 2024

Accepted 17 January 2025

Available online 27 January 2025

Keywords:

Digraph packing

Perfect out-forest

Directed Steiner cycle packing

Semicomplete digraph

Acyclic digraph

Eulerian digraph

Symmetric digraph

ABSTRACT

The perfect out-forest problem generalizes the perfect matching problem, and the directed Steiner cycle packing problem generalizes the Hamiltonian cycle decomposition problem and is a variant of the directed Steiner tree packing problem. In this paper, we study the complexity of these two types of digraph packing problems.

For the perfect out-forest problem, Gutin and Yeo proved that it is NP-complete to decide whether a given strong digraph contains a 0-perfect out-forest. We show that this result also holds for the 1-perfect out-forest problem. However, when restricted to a semicomplete digraph D , the problem of deciding whether D contains an i -perfect out-forest becomes polynomial-time solvable, where $i \in \{0, 1\}$. In addition, we prove that it is NP-hard to find a 0-perfect out-forest of maximum size in a connected acyclic digraph, and it is NP-hard to find a 1-perfect out-forest of maximum size in a connected digraph.

For the directed Steiner cycle packing problem, when both $k \geq 2, \ell \geq 1$ are fixed integers, we show that the problem of deciding whether there are at least ℓ internally disjoint directed S -Steiner cycles in a digraph D is NP-complete, where $S \subseteq V(D)$ and $|S| = k$. However, when we consider the class of symmetric digraphs, the problem becomes polynomial-time solvable. We also show that the problem of deciding whether there are at least ℓ arc-disjoint directed S -Steiner cycles in a given digraph D is NP-complete, where $S \subseteq V(D)$ and $|S| = k$.

© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

1.1. Motivation, terminology and notation

We refer the readers to [1] for graph-theoretical notation and terminology not given here. Note that all digraphs considered in this paper have no parallel arcs or loops. We use $[n]$ to denote the set of all natural numbers from 1 to n . A digraph D is called *connected* if $UG(D)$ is connected, where $UG(D)$ is the underlying undirected graph of D . Moreover, D is *strongly connected* or, simply, *strong*, if for any pair of vertices $x, y \in V(D)$, there is a path from x to y in D , and vice versa. A digraph D is *semicomplete* if for every pair of distinct vertices $x, y \in V(D)$ at least one of the arcs xy, yx is in D . A *tournament* is a semicomplete digraph without 2-cycles. A digraph D is *symmetric* if every arc in D belongs to a 2-cycle. A digraph D is *Eulerian* if D is connected and $d^+(x) = d^-(x)$ for every vertex $x \in V(D)$. A digraph is *acyclic* if it has no directed cycle. An *out-tree* (resp. *in-tree*) *rooted at a vertex r* is an orientation of a tree such that the in-degree (resp. out-degree)

* Corresponding author.

E-mail addresses: sunyuefang@nbu.edu.cn (Y. Sun), zeminjin@zjnu.cn (Z. Jin).

of every vertex but r equals one. An *out-branching* B_r^+ (resp. *in-branching* B_r^-) in a digraph D is a spanning subdigraph of D which is an out-tree (resp. in-tree).

For $i \in \{0, 1\}$, a spanning forest F of an undirected graph G is an *i-perfect forest* if each tree of F is an induced subgraph of G , and exactly i vertices in G have even degrees (including zero). A 0-perfect forest is also called a *perfect forest*. Clearly, the concept of perfect forest is a natural generalization of perfect matching of graphs. The problem of *i*-perfect forest in undirected graphs has been studied by some researchers [2,6,8,14,15].

Gutin and Yeo [7] introduced the concept of perfect out-forest in digraphs. An *out-forest* is a collection of vertex disjoint out-trees. A spanning out-forest F of a digraph D is a *perfect out-forest* if each out-tree of F is an induced subgraph of D , and the degree of each vertex in $UG(F)$ is odd. Clearly, the concept of perfect out-forest is a natural generalization of perfect matching of digraphs. Similarly, we now define the concept of *i*-perfect out-forest as follows. For $i \in \{0, 1\}$, a spanning out-forest F of a digraph D is an *i-perfect out-forest* if each out-tree of F is an induced subgraph of D , and exactly i vertices in $UG(F)$ have even degrees (including zero). Clearly, a 0-perfect out-forest is exactly a perfect out-forest.

The basic problem of STEINER TREE PACKING (also called EDGE-DISJOINT STEINER TREE PACKING PROBLEM) is defined as follows: the input consists of an undirected graph G and a subset of vertices $S \subseteq V(D)$, the goal is to find a largest collection of edge-disjoint S -Steiner trees (an S -Steiner tree is a tree T with $S \subseteq V(T)$). The Steiner tree packing problem has applications in VLSI circuit design [4,5,16] and attracts much attention from researchers. Besides the classical version, people also study some other variations, such as internally disjoint Steiner tree packing problem [3,10]. It is natural to extend Steiner tree packing problem to digraphs, and such extensions are called directed Steiner type packing problems, including directed Steiner tree packing problem [3,21] and strong subgraph packing problem [18–20].

Now we introduce a new Steiner type packing problem in digraphs, we call it the directed Steiner cycle packing problem. Let $D = (V(D), A(D))$ be a digraph of order n , $S \subseteq V(D)$ a k -subset of $V(D)$, where $2 \leq k \leq n$. A directed cycle C of D is called a *directed S-Steiner cycle* or, simply, an *S-cycle* if $S \subseteq V(C)$. Two S -cycles are said to be *arc-disjoint* if they have no common arc. Furthermore, two arc-disjoint S -cycles are *internally disjoint* if the set of common vertices of them is exactly S . Let *internally (resp. arc-)disjoint cycle packing number*, denoted by $\kappa_S^c(D)$ (resp. $\lambda_S^c(D)$), be the maximum number of internally (resp. arc-)disjoint S -cycles in D . We define the following two types of directed Steiner cycle packing problems:

ARC-DISJOINT DIRECTED STEINER CYCLE PACKING: The input consists of a digraph D and a subset of vertices $S \subseteq V(D)$, the goal is to find a largest collection of arc-disjoint S -cycles.

INTERNALLY DISJOINT DIRECTED STEINER CYCLE PACKING: The input consists of a digraph D and a subset of vertices $S \subseteq V(D)$, the goal is to find a largest collection of internally disjoint S -cycles.

It is worth noting that Steiner cycles have applications in the optimal design of reliable telecommunication and transportation networks [17]. And directed Steiner cycle packing problems are highly related to other problems in graph theory. For example, when $|S| = n$, an S -cycle is a Hamiltonian cycle. Therefore, in this situation, these two problems are equivalent to finding the maximum number of arc-disjoint Hamiltonian cycles, and so generalize the Hamiltonian cycle decomposition problem. A digraph D is *k-cyclic* if it has a cycle containing the vertices $x_1, x_2, \dots, x_k$ for every choice of k vertices [9]. By definition, a digraph is *k-cyclic* if and only if $\kappa_S^c(D) \geq 1$ (resp. $\lambda_S^c(D) \geq 1$) for every k -subset S of $V(D)$.

1.2. Our results

In Section 2, by using the NP-completeness for the problem of perfect out-forest in strong digraphs [7], we prove the NP-completeness of deciding whether a given strong digraph contains a 1-perfect out-forest (Theorem 2.2(a)). However, when restricted to semicomplete digraphs, the problem of deciding whether D contains an *i*-perfect out-forest becomes polynomial-time solvable, where $i \in \{0, 1\}$ (Theorem 2.2(b)). The problem of finding a perfect out-forest of size at least $|V(D)| - 1$ in a digraph D is polynomial-time solvable because D has a perfect forest of size at least $|V(D)| - 1$ if and only if D is an out-branching in which every vertex is of odd degree in $UG(D)$. However, as shown in Theorem 2.3, when “ $|V(D)| - 1$ ” is replaced by “ $|V(D)| - k$ ” ($k \geq 2$), the problem above becomes NP-hard, even when D is a connected acyclic digraph. Similarly, we also prove that it is NP-hard to find a 1-perfect out-forest of size at least $|V(D)| - k$ in a connected digraph D for a fixed integer $k \geq 2$ (Theorem 2.4). These two theorems imply that it is NP-hard to find a perfect out-forest of maximum size in a connected acyclic digraph, and it is NP-hard to find a 1-perfect out-forest of maximum size in a connected digraph (Theorem 2.5).

In Section 3, we study the problem of Steiner cycle packing. When both $k \geq 2, \ell \geq 1$ are fixed integers, we show that the problem of deciding whether $\kappa_S^c(D) \geq \ell$ with $|S| = k$ is NP-complete (Theorem 3.2). However, when we consider the class of symmetric digraphs, the problem becomes polynomial-time solvable (Theorem 3.6). We also show that the problem of deciding whether $\lambda_S^c(D) \geq \ell$ with $|S| = k$ is NP-complete, when both $k \geq 2, \ell \geq 1$ are fixed integers (Theorem 3.3).

2. Perfect out-forest problem

Gutin and Yeo proved the complexity for the problem of deciding whether a given strong digraph D contains a perfect out-forest.

Theorem 2.1 ([7]). *The problem of deciding whether a given strong digraph D contains a perfect out-forest is NP-complete.*

By [Theorem 2.1](#), we can prove a similar result for the problem of 1-perfect out-forest. These two results mean that the problem of i -perfect out-forest ($i \in \{0, 1\}$) in a general digraph is difficult. However, when restricted to some digraph classes, such as semicomplete digraphs, the problem becomes polynomial-time solvable. Especially, for a tournament T with order n , it is not hard to show that T contains a perfect out-forest if and only if n is even, and T contains a 1-perfect out-forest if and only if n is odd.

Theorem 2.2. *The following assertions hold:*

- (a) *The problem of deciding whether a given strong digraph D contains a 1-perfect out-forest is NP-complete.*
- (b) *The problem of deciding whether a given semicomplete digraph D contains an i -perfect out-forest is polynomial-time solvable, where $i \in \{0, 1\}$.*

Proof. Part (a): It is not difficult to see that the problem belongs to NP, so it suffices to show that the problem is NP-hard. Let D be any strong digraph, we construct a strong digraph D' from D by adding a new vertex v and two arcs uv, vu for some vertex $u \in V(D)$. If D contains a perfect out-forest F , then $F' = F \cup \{v\}$ is clearly a 1-perfect out-forest of D' such that v is the unique vertex whose degree in the underlying graph of F' is even. If D' contains a 1-perfect out-forest F' , then $\{v\} \in F'$. Indeed, if $\{v\} \notin F'$, then there is an out-tree $T \in F'$ containing u, v . As T is an induced subgraph of D , we have $uv, vu \in A(T)$, a contradiction. Hence $\{v\} \in F'$ and v is the unique vertex whose degree in the underlying graph of F' is even. Now observe that $F = F' \setminus \{v\}$ is a perfect out-forest of D . Therefore, D contains a perfect out-forest if and only if D' contains a 1-perfect out-forest, by [Theorem 2.1](#), the result holds.

Part (b): Let D be a semicomplete digraph. Observe that F is a perfect out-forest of D if and only if F is a perfect matching of D such that if $xy \in A(F)$, then $yx \notin A(D)$. Now we remove all arcs that belong to 2-cycles in D . That is, if xy and yx are both arcs in D , then neither of them can be used in the matching F , so delete both of them from the digraph. Then the problem is equivalent to deciding if the resulting digraph (with no 2-cycles), denoted by D' , has a perfect matching. Clearly, this problem is polynomial-time solvable, as D' has a perfect matching if and only if $UG(D')$ has a perfect matching, and the latter can be decided in polynomial-time. Hence, the problem of deciding whether a given semicomplete digraph D contains a perfect out-forest is polynomial-time solvable.

Observe that D has a 1-perfect out-forest if and only if for some vertex $u \in V(D)$, $D - u$ has a perfect matching F' such that if $xy \in A(F')$, then $yx \notin A(D - u)$. By the argument above, we can in polynomial-time decide if $D - u$ has such a perfect matching. Hence, the problem of deciding whether a given semicomplete digraph D contains a 1-perfect out-forest is also polynomial-time solvable. $\square$

Recall that the problem of finding a perfect out-forest of size at least $|V(D)| - 1$ in a digraph D is polynomial-time solvable. However, as shown in [Theorem 2.3](#), when “ $|V(D)| - 1$ ” is replaced by “ $|V(D)| - k$ ” ($k \geq 2$), the problem above becomes NP-complete, even when D is a connected acyclic digraph. The *not-all-equal 3-SAT* problem, abbreviated to NAE-3-SAT, is the problem of determining whether an instance of 3-SAT has a truth assignment to its variables such that each clause contains both a true and a false literal. If such case holds for an instance I , then we say that I is *NAE-satisfied*.

Theorem 2.3. *Let $k \geq 2$ be an integer. The problem of deciding whether a given connected acyclic digraph D contains a perfect out-forest with at least $|V(D)| - k$ arcs is NP-complete.*

Proof. It is not difficult to see that the problem belongs to NP, so it suffices to show that the problem is NP-hard. We first consider the case that $k = 2$, that is, we show that the problem of deciding whether a given connected acyclic digraph D contains a perfect out-forest with at least $|V(D)| - 2$ arcs is NP-hard, by reducing from the NAE-3-SAT problem which is NP-hard [12].

Let I be an instance of NAE-3-SAT with clauses $C_1, C_2, \dots, C_m$ and variables $v_1, v_2, \dots, v_n$. Our construction is based on a construction from [8]. For $i \in [n]$, let H_i be a digraph (see [Fig. 1](#)) with vertex set

$$\{x_1^i, z_1^i, y_1^i, x_2^i, z_2^i, y_2^i\}$$

and arc set

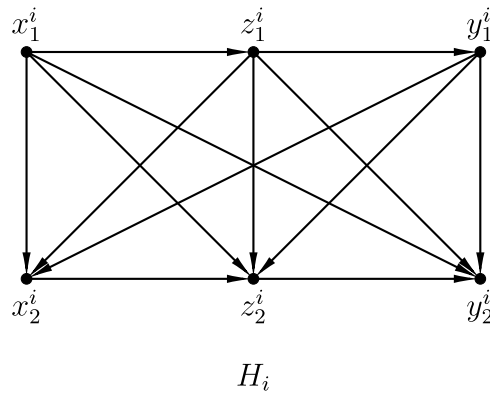
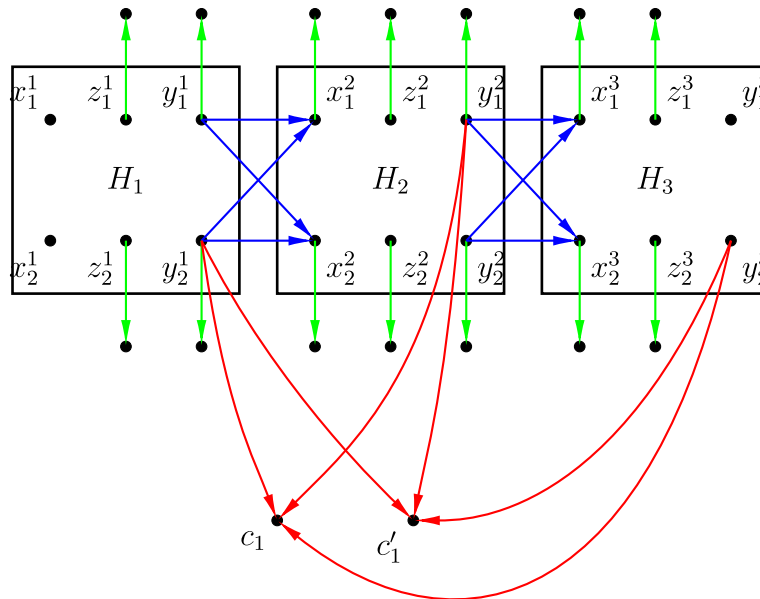
$$\{x_1^i z_1^i, z_1^i y_1^i, x_2^i z_2^i, z_2^i y_2^i, x_1^i x_2^i, x_1^i z_2^i, x_1^i y_2^i, z_1^i x_2^i, z_1^i z_2^i, z_1^i y_2^i, y_1^i x_2^i, y_1^i z_2^i, y_1^i y_2^i\}.$$

For $i \in [n - 1]$, we add all arcs from $\{y_1^i, y_2^i\}$ to $\{x_1^{i+1}, x_2^{i+1}\}$: $y_1^i x_1^{i+1}, y_2^i x_1^{i+1}, y_1^i x_2^{i+1}, y_2^i x_2^{i+1}$. For each vertex $u \in V(H_i) \setminus \{x_1^i, x_2^i, y_1^i, y_2^i\}$ ($i \in [n]$), add a new vertex u' and a new arc uu' . Now the current digraph is denoted by Q .

Let $V(D) = V(Q) \cup \{c_j, c'_j \mid j \in [m]\}$. For each $j \in [m]$, we add the arcs $y_2^j c_j, y_2^j c'_j$ (resp. $y_1^j c_j, y_1^j c'_j$) if and only if v_i (resp. $\bar{v}_i$) is a literal in the clause C_j . This completes the construction of D . Observe that D is a connected acyclic digraph. See [Fig. 2](#) for an example of D when $I = (v_1, \bar{v}_2, v_3)$.

To complete the proof of the case that $k = 2$, it suffices to show that D contains a perfect out-forest of size at least $|V(D)| - 2$ if and only if I is NAE-satisfied.

We first assume that D contains a perfect out-forest F with at least $|V(D)| - 2$ arcs. Observe that now F consists of two vertex-disjoint out-trees T_1 and T_2 such that each T_i is an induced subdigraph in D of order at least two, the degree of each vertex in $UG(T_i)$ is odd, and $V(T_1) \cup V(T_2) = V(D)$. We will prove the following three claims, and complete the proof of this direction in [Claim 3](#).

Fig. 1. The digraph H_i .Fig. 2. The digraph D if $I = (v_1, \bar{v}_2, v_3)$.

Claim 1. For each $i \in [n]$, exactly one of the following assertions hold:

- A: $\{x_1^i, z_1^i, y_1^i\} \subseteq V(T_1)$ and $\{x_2^i, z_2^i, y_2^i\} \subseteq V(T_2)$.
- B: $\{x_1^i, z_1^i, y_1^i\} \subseteq V(T_2)$ and $\{x_2^i, z_2^i, y_2^i\} \subseteq V(T_1)$.

Proof of Claim 1. Observe that $|V(T_j) \cap V(H_i)| = 3$ for $j \in [2]$ and $i \in [n]$. Indeed, if $|V(T_j) \cap V(H_i)| \geq 4$ for some pair (i, j) with $i \in [n]$ and $j \in [2]$, then the underlying graph of $D[V(Q) \cap V(T_j)]$ contains a 4-cycle, a contradiction. By the fact that the in-degree of each vertex (except the root vertex) in an out-tree is one, x_1^i, z_1^i, y_1^i must belong to the same tree, say T_j , and so x_2^i, z_2^i, y_2^i must belong to another tree T_{3-j} . This completes the proof of Claim 1. ■

Claim 2. $|N_D(c_j) \cap V(T_i)| = 1$ whenever $c_j \in V(T_i)$.

Proof of Claim 2. Recall that both T_1 and T_2 have orders at least two, any vertex having degree one in $UG(D)$ must belong to the same tree as its neighbour. By Claim 1, $D[V(Q) \cap V(T_i)]$ is a directed path of length $3n$ with a pendent out-arc from each inner vertex of the path.

As both T_1 and T_2 have orders at least two, we have $|N_D(c_j) \cap V(T_i)| \geq 1$. Suppose that $|N_D(c_j) \cap V(T_i)| \geq 2$. As T_i is an induced out-tree in Q , c_j must have at least two neighbours, say x and y , in T_i . By the argument above, there is an $x - y$ path in the underlying graph of $D[V(Q) \cap V(T_i)]$, then together with the vertex c_j , there is a cycle in $UG(T_i)$, a contradiction. This completes the proof of Claim 2. ■

Claim 3. *The instance I is NAE-satisfiable.*

Proof of Claim 3. We give a truth assignment as follows: if $y_1^i \in V(T_1)$, then let v_i be true, and if $y_2^i \in V(T_1)$ (which means that $y_1^i \notin V(T_1)$), then let v_i be false.

Without loss of generality, we assume that $c_j \in V(T_1)$ (the argument for the case that $c_j \in V(T_2)$ is similar and hence we omit the details), and the two variables v_{i_1}, v_{i_2} appear in C_j . We first consider the case that $y_1^{i_1} \in V(T_1)$. Now $c_j y_1^{i_1} \in A(T_1)$, so $\overline{v_{i_1}}$ is a literal of C_j (by the construction of D). As now v_{i_1} is true, the literal $\overline{v_{i_1}}$ is false. By Claim 2, we have $|N_D(c_j) \cap V(T_1)| = 1$, so $c_j y_1^{i_2} \in A(T_2)$ or $c_j y_2^{i_2} \in A(T_2)$. For the former case, $\overline{v_{i_2}}$ is a literal of C_j and $y_2^{i_2} \in V(T_1)$ (and so v_{i_2} is false), then $\overline{v_{i_2}}$ is a true literal of C_j . For the latter case, v_{i_2} is a literal of C_j and $y_1^{i_2} \in V(T_1)$ (and so v_{i_2} is true), then v_{i_2} is still a true literal of C_j . Hence, we have that C_j has at least one true literal and at least one false literal in this case. For the case that $y_2^{i_1} \in V(T_1)$, the argument is similar, so we omit the details. This completes the proof of Claim 3 and the first direction. ■

Conversely, assume that I is NAE-satisfied and consider a truth assignment τ NAE-satisfying I . We now construct two vertex-disjoint induced out-trees, say T_1 and T_2 , in D such that $V(T_1) \cup V(T_2) = V(D)$, and all degrees in $UG(T_i)$ are odd for $i \in [2]$. If v_i is true (resp. false) in τ , then add vertices x_1^i, z_1^i, y_1^i (resp. x_2^i, z_2^i, y_2^i) to T_1 and add vertices x_2^i, z_2^i, y_2^i (resp. x_1^i, z_1^i, y_1^i) to T_2 . We then update T_i by adding u' (if exists) for each vertex u of the current T_i ($i \in [2]$). Note that now each $V(T_i)$ induces an out-tree in D , and each vertex of $V(T_i)$ has odd degree in $UG(T_i)$. We still need the following claim.

Claim 4. *Each of c_j and c'_j has exactly one arc coming from one tree and two arcs coming from the other tree, where $j \in [m]$.*

Proof of Claim 4. Suppose that all three arcs incident to c_j (resp. c'_j) come from T_1 , and the variables $v_{i_1}, v_{i_2}, v_{i_3}$ appear in C_j . Furthermore, without loss of generality, assume that $y_1^{i_1}, y_1^{i_2}, y_1^{i_3} \in V(T_1)$. This means that C_j consists of the following three literals: $\overline{v_{i_1}}, \overline{v_{i_2}}, \overline{v_{i_3}}$, and all of $v_{i_1}, v_{i_2}, v_{i_3}$ are true, which produces a contradiction as now C_j contains no true literal. This completes the proof of Claim 4. ■

In fact, we can further prove that the tree which has exactly one arc into c_j is the same as the tree which has exactly one arc into c'_j by the construction of D . Now add each of c_j and c'_j to the tree from which there is exactly one arc. It can be checked that T_1 and T_2 form a perfect out-forest in D with $|V(D)| - 2$ arcs, where T_i is an induced out-tree rooted at x_i^1 for $i \in [2]$. This completes the proof for the other direction and the case that $k = 2$.

We now prove the case that $k \geq 3$. Let D be a connected acyclic digraph, and let D' be obtained from D by adding $2k - 4$ new vertices: u_i ($1 \leq i \leq 2k - 4$), such that

$$A(D') = A(D) \cup \{u_{i+1}u_i \mid 0 \leq i \leq 2k - 5\},$$

where $u_0 \in V(D)$. Clearly, D' is also connected and acyclic. If D has a perfect out-forest, say F , of size at least $|V(D)| - 2$, then $F' = F \cup \{u_{2\ell}u_{2\ell-1} \mid 1 \leq \ell \leq k - 2\}$ is a perfect out-forest in D' of size at least $|V(D)| - 2 + (k - 2) = |V(D')| - k$. Conversely, if D' has a perfect out-forest, say F' , of size at least $|V(D')| - k$, then each $u_{2\ell}u_{2\ell-1} \in F'$ by the degree of each u_i and the definition of a perfect out-forest. Now $F = F' \setminus \{u_{2\ell}u_{2\ell-1} \mid 1 \leq \ell \leq k - 2\}$ is a perfect out-forest in D of size at least $|V(D')| - k - (k - 2) = |V(D)| - 2$. Therefore, D has a perfect out-forest of size at least $|V(D)| - 2$ if and only if D' has a perfect out-forest of size at least $|V(D')| - k$ where $k \geq 3$. This completes the proof of the case that $k \geq 3$. □

Similarly, the problem of finding a 1-perfect out-forest of size at least $|V(D)| - 1$ in a digraph D is also polynomial-time solvable because D has a 1-perfect forest of size at least $|V(D)| - 1$ if and only if D is an out-branching in which exactly one vertex is of even degree in $UG(D)$. However, as shown in Theorem 2.4, when “ $|V(D)| - 1$ ” is replaced by “ $|V(D)| - k$ ” ($k \geq 2$), the problem above becomes NP-complete.

Theorem 2.4. *Let $k \geq 2$ be an integer. The problem of deciding whether a given connected digraph D contains a 1-perfect out-forest with at least $|V(D)| - k$ arcs is NP-complete.*

Proof. It is not difficult to see that the problem belongs to NP, so it suffices to show that the problem is NP-hard. We first consider the case that $k = 2$. Let D' be a connected digraph constructed from the digraph D in Theorem 2.3 such that $V(D') = V(D) \cup \{x\}$ and $A(D') = A(D) \cup \{xx^1\}$. With a similar argument to that of Theorem 2.3, we can show that D' contains a 1-perfect forest of size at least $|V(D')| - 2$ if and only if I is NAE-satisfied. By the NP-hardness of NAE 3-SAT problem, this case holds.

We next consider the case that $k \geq 3$. Let Q be a connected digraph, we construct another connected digraph Q' from Q by adding a new vertex v and two arcs uv, vu for some vertex $u \in V(Q)$. If Q contains a perfect out-forest F of size $|V(Q)| - k + 1$, then F consists of $k - 1$ disjoint out-trees, that is, $F = \{T_i \mid i \in [k - 1]\}$. So $F' = F \cup \{v\}$ is a 1-perfect out-forest in Q' of size $|V(Q')| - k$, where v is the unique vertex whose degree in the underlying graph of F' is even. If Q' contains a 1-perfect out-forest F' of size $|V(Q')| - k$, then $\{v\} \in F'$. Indeed, if $\{v\} \notin F'$, then there is an out-tree $T \in F'$ containing u, v . As T is an induced subgraph of Q , we have $uv, vu \in A(T)$, a contradiction. Hence $\{v\} \in F'$ and v is the unique vertex whose degree in the underlying graph of F' is even. Now observe that $F = F' \setminus \{v\}$ is a perfect out-forest in Q of size $|V(Q)| - k + 1$. Therefore, Q contains a perfect out-forest of size $|V(Q)| - k + 1$ if and only if Q' contains a 1-perfect out-forest of size $|V(Q')| - k$, by Theorem 2.3, this case also holds. □

Theorems 2.3 and 2.4 directly imply the following result.

Theorem 2.5. *The following assertions hold:*

- (a) *It is NP-hard to find a perfect out-forest of maximum size in a connected acyclic digraph.*
- (b) *It is NP-hard to find an 1-perfect out-forest of maximum size in a connected digraph.*

However, when restricted to some other digraph classes, such as semicomplete digraphs, the above two problems become polynomial-time solvable. As shown in the proof of Theorem 2.2(b), F is a perfect out-forest of a semicomplete digraph D if and only if F is a perfect matching of D such that if $xy \in A(F)$, then $yx \notin A(D)$. Hence, each perfect out-forest (if exists) of D has $|V(D)|/2$ arcs and we can in polynomial-time find a perfect out-forest (if exists) of maximum size in a semicomplete digraph according to the argument of Theorem 2.2(b). Similarly, we can also in polynomial-time find a 1-perfect out-forest (if exists) of maximum size in a semicomplete digraph.

3. Steiner cycle packing problem

The problem of DIRECTED k -LINKAGE is defined as follows: Given a digraph D and a (terminal) sequence $(s_1, t_1, \dots, s_k, t_k)$ of distinct vertices of D , decide whether D has k vertex-disjoint paths $P_1, \dots, P_k$, where P_i starts at s_i and ends at t_i for all $i \in [k]$. Sun and Yeo proved the following NP-completeness of DIRECTED 2-LINKAGE for Eulerian digraphs.

Theorem 3.1 ([21]). *The problem of DIRECTED 2-LINKAGE restricted to Eulerian digraphs is NP-complete.*

Using Theorem 3.1, we can prove the following NP-completeness of deciding whether $\kappa_S^c(D) \geq \ell$ for digraphs.

Theorem 3.2. *Let $k \geq 2, \ell \geq 1$ be fixed integers. It is NP-complete to decide, given a digraph D and $S \subseteq V(D)$ with $|S| = k$, whether $\kappa_S^c(D) \geq \ell$.*

Proof. It is not difficult to see that the problem belongs to NP. We will show that the problem is NP-hard by reducing from DIRECTED 2-LINKAGE in Eulerian digraphs. Let $[H; s_1, s_2, t_1, t_2]$ be an instance of DIRECTED 2-LINKAGE in Eulerian digraphs, that is, H is an Eulerian digraph, and (s_1, t_1, s_2, t_2) is a (terminal) sequence of distinct vertices of H .

We just consider the case that $k \geq 2, \ell \geq 2$ as the argument for the remaining case is similar and simpler. We produce a new digraph D as follows. Let

$$V(H') = V(H) \cup S \cup \{r_1, r_2\},$$

where $S = \{x_i \mid i \in [k]\}$, and let

$$\begin{aligned} A(H') = & A(H) \cup \{x_{k-1}s_1, t_1x_k, x_k s_2, t_2x_1, s_1r_1, r_1t_2, s_2r_2, r_2t_1\} \\ & \cup \{x_i x_{i+1} \mid i \in [k]\}, \end{aligned}$$

where $x_{k+1} = x_1$. Furthermore, we replicate the arc $x_i x_{i+1}$ $\ell - 1$ additional copies for each $i \in [k - 2]$ and replicate the two arcs $x_{k-1}x_k$ and $x_k x_1$ $\ell - 2$ additional copies (note that when $k = 2$, we just replicate the two arcs $x_1 x_2$ and $x_2 x_1$ $\ell - 2$ additional copies). Finally, to avoid parallel arcs, insert a new vertex $z_{i,i+1}^j$ to each arc $x_i x_{i+1}$, where $j \in [\ell]$ if $i \in [k - 2]$ and $j \in [\ell - 1]$ otherwise. The resulting digraph is called D as shown in Fig. 3.

We will show that $\kappa_S^c(D) \geq \ell$ if and only if there exist two vertex-disjoint paths, P_1 and P_2 , in H such that P_i is an $s_i - t_i$ path, for $i \in [2]$. This will complete the proof of this case by Theorem 3.1.

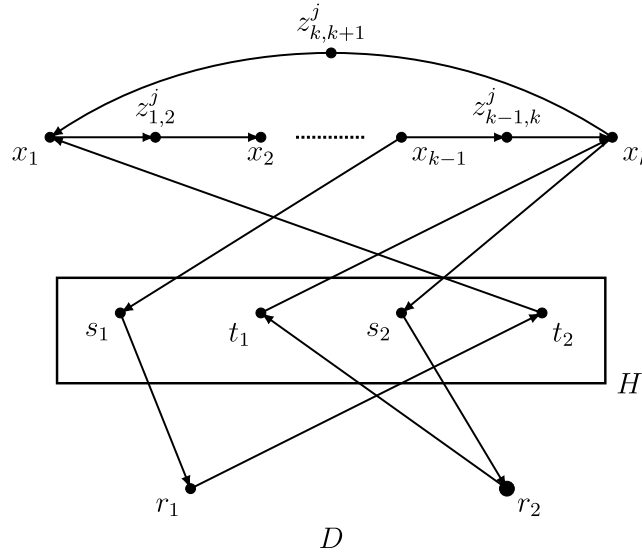
First assume that there exist two vertex-disjoint paths, P_1 and P_2 in H , such that P_i is an $s_i - t_i$ path, for $i \in [2]$. Add the arcs $x_{k-1}s_1, t_1x_k, x_k s_2, t_2x_1$, and the path $x_1 z_{1,2}^\ell x_2 \dots z_{k-2,k-1}^\ell x_{k-1}$ to P_1, P_2 and call the resulting cycle for C_ℓ . Let C_j be the cycle $x_1 z_{1,2}^j x_2 \dots z_{k-1,k}^j x_k z_{k,k+1}^j x_1$, for each $j \in [\ell - 1]$. It can be checked that the above cycles $C_1, C_2, \dots, C_\ell$ are pairwise internally disjoint, which implies that $\kappa_S^c(D) \geq \ell$ as desired.

Conversely, assume that $\kappa_S^c(D) \geq \ell$, that is, there is a set of pairwise internally disjoint S -cycles, say $\{C_i \mid i \in [\ell]\}$. Since $\deg_D^+(x_i) = \deg_D^-(x_i) = \ell$ for every $i \in [\ell]$, each C_i contains an $x_k - x_1$ path, an $x_1 - x_{k-1}$ path and an $x_{k-1} - x_k$ path. And there is an $x_{k-1} - x_k$ path, say P_i , containing s_1, t_1 in C_i , and there is an $x_k - x_1$ path, say P_j , containing s_2, t_2 in C_j .

If $i = j$, then clearly there is a pair of vertex-disjoint $s_1 - t_1$ path and $s_2 - t_2$ path in the cycle C_i (and therefore in H), as desired. It remains to consider the case that $i \neq j$. Without loss of generality, assume that $i = 1, j = 2$. Observe that there is an $x_k - x_{k-1}$ path P'_1 in C_1 and an $x_{k-1} - x_k$ path P'_2 in C_2 , that is, $C_i = P_i \cup P'_i$ for each $i \in [2]$. Let $C'_1 = P_1 \cup P_2$ and $C'_2 = P'_1 \cup P'_2$. It can be checked that $\{C'_i, C_j \mid 1 \leq i \leq 2, 3 \leq j \leq \ell\}$ is a set of internally disjoint S -cycles. With a similar argument to that of the above case, we still deduce that there is pair of vertex-disjoint $s_1 - t_1$ path and $s_2 - t_2$ path in the cycle C_1 (and therefore in H). $\square$

We can also prove the NP-completeness of deciding whether $\lambda_S^c(D) \geq \ell$ for general digraphs.

Theorem 3.3. *Let $k \geq 2, \ell \geq 1$ be fixed integers. It is NP-complete to decide, given a digraph D and $S \subseteq V(D)$ with $|S| = k$, whether $\lambda_S^c(D) \geq \ell$.*

Fig. 3. The digraph D .

Proof. When $k \geq 2$, $\ell = 1$, by definitions, we clearly have $\lambda_S^\ell(D) \geq 1$ if and only if $\kappa_S^\ell(D) \geq 1$, so the result holds for this case by Theorem 3.2. Now in this following argument, we assume that $k \geq 2$, $\ell \geq 2$.

It is not difficult to see that the problem belongs to NP. We will show that the problem is NP-hard by reducing from DIRECTED 2-LINKAGE in Eulerian digraphs. Let $[H; s_1, s_2, t_1, t_2]$ be an instance of DIRECTED 2-LINKAGE in Eulerian digraphs, that is, H is an Eulerian digraph, and (s_1, t_1, s_2, t_2) is a (terminal) sequence of distinct vertices of H .

We first produce a digraph D the same as that in Theorem 3.2. Secondly, we construct a new digraph D' from D using a classical reduction (see (70.7) in [13]) as follows: replace every vertex u of H by two vertices u^- and u^+ such that u^-u^+ is an arc in D' and for every $uv \in A(H)$ add an arc u^+v^- to D' . Also, for $z \in S \cup \{r_1, r_2\}$, for every arc zu in D add an arc zu^- to D' and for every arc uz add an arc u^+z to D' .

It was proved in Theorem 3.2 that $\kappa_S^\ell(D) \geq \ell$ if and only if $[H; s_1, s_2, t_1, t_2]$ is a positive instance of DIRECTED 2-LINKAGE in Eulerian digraphs. Therefore, to prove the theorem, it suffices to show that $\lambda_S^\ell(D') \geq \ell$ if and only if $\kappa_S^\ell(D) \geq \ell$.

We first assume that $\kappa_S^\ell(D) \geq \ell$, that is, there is a set of ℓ internally disjoint S -cycles in D , say $\{C_i \mid i \in [\ell]\}$. By the construction of D , we have $\deg_D^+(x_i) = \ell$, each C_i contains precisely one out-neighbour and one in-neighbour of x_i . There must be two paths, say P_1, P_2 , such that P_1 contains s_1, t_1 (resp. P_2 contains s_2, t_2), and starts from x_{k-1} (resp. x_k) and ends at x_k (resp. x_{k-1}) in C_i (resp. C_j) for some $i \in [\ell]$ (resp. $j \in [\ell]$). Now we obtain a cycle C'_i (resp. C'_j) in D' from C_i (resp. C_j) as follows: replace every vertex u of $V(H) \cap V(C_i)$ (resp. $V(H) \cap V(C_j)$) by two vertices u^- and u^+ such that u^-u^+ is an arc in C'_i (resp. C'_j) and for every $uv \in A(H) \cap A(C_i)$ (resp. $uv \in A(H) \cap A(C_j)$) add an arc u^+v^- to $A(C'_i)$ (resp. $A(C'_j)$). Also, for each $z \in S \cup \{r_1, r_2\}$, for every arc zu in $A(D) \cap A(C_i)$ (resp. $A(D) \cap A(C_j)$) add an arc zu^- to C'_i (resp. C'_j), and for every arc uz add an arc u^+z to C'_i (resp. C'_j). Then combining with the remaining $\ell - 2$ S -cycles in $\{C_i \mid i \in [\ell]\}$, we obtain a set of ℓ arc-disjoint S -cycles in D' , therefore $\lambda_S^\ell(D') \geq \ell$.

Conversely, we assume that $\lambda_S^\ell(D') \geq \ell$, that is, there is a set of ℓ arc-disjoint S -cycles in D' , say $\{C_i \mid i \in [\ell]\}$. By the construction of D' , we have $\deg_{D'}^+(x_i) = \ell$, each C_i contains precisely one out-neighbour and one in-neighbour of x_i . There must be two paths, say Q_1, Q_2 , such that Q_1 contains s_1^-, t_1^+ (resp. Q_2 contains s_2^-, t_2^+), and starts from x_{k-1} (resp. x_k) and ends at x_k (resp. x_{k-1}) in C_i (resp. C_j) for some $i \in [\ell]$ (resp. $j \in [\ell]$). Observe that Q_1 must be of the form $x_{k-1}s_1^-s_1^+a_1^+ \dots a_p^+t_1^+t_1^-x_k$ and Q_2 must be of the form $x_k s_2^-s_2^+b_1^+ \dots b_q^+t_2^+t_2^-x_{k-1}$ and furthermore, we have $\{a_i \mid i \in [p]\} \cap \{b_j \mid j \in [q]\} = \emptyset$. Now we obtain C'_i (resp. C'_j) from C_i (resp. C_j) by replacing Q_1 (resp. Q_2) with the path $x_{k-1}s_1a_1 \dots a_pt_1x_k$ (resp. $x_k s_2b_1 \dots b_qt_2x_{k-1}$). It can be checked that, combining with the remaining $\ell - 2$ S -cycles in $\{C_i \mid i \in [\ell]\}$, we get a set of ℓ internally disjoint S -cycles in D , therefore $\kappa_S^\ell(D) \geq \ell$. $\square$

Recall that in Theorem 3.2, we showed that it is NP-complete to decide, given any digraph D and $S \subseteq V(D)$ with $|S| = k$, whether $\kappa_S^\ell(D) \geq \ell$, where $k \geq 2$, $\ell \geq 1$ are both fixed integers. However, when we consider the class of symmetric digraphs, the problem becomes polynomial-time solvable.

The definition of the UNDIRECTED p -LINKAGE problem is given below:

UNDIRECTED p -LINKAGE: For a fixed integer $p \geq 2$, given a graph G and a (terminal) sequence $((s_1, t_1), \dots, (s_p, t_p))$ of distinct vertices of G , decide whether G has p vertex-disjoint paths $P_1, \dots, P_p$, where P_i connects s_i and t_i for all $i \in [p]$.

Robertson and Seymour proved:

Theorem 3.4 ([11]). *There is an algorithm that runs in time $f(p)n^3$ for the UNDIRECTED p -LINKAGE problem for some function $f(p)$ of the parameter p .*

Theorem 3.4 means that there is an algorithm in time $O(n^3)$ for the UNDIRECTED p -LINKAGE problem when p is fixed. The following result is a consequence of Theorem 3.4.

Lemma 3.5 ([21]). *Let D be a symmetric digraph with $S \subseteq V(D)$. Let $\{s_i, t_i \mid i \in [p]\} \subseteq V(D)$. The following problem can be solved in time $f(p)|V(D)|^3$ for some function $f(p)$: decide whether there is an (s_i, t_i) -path, say P_i , for each $i \in [p]$, such that none of the internal vertices of P_i are contained in $S \cup \bigcup_{j \in [p] \setminus \{i\}} V(P_j)$.*

By Lemma 3.5, we will now prove the polynomiality for $\kappa_S^c(D)$ on symmetric digraphs.

Theorem 3.6. *Let $k \geq 2$ and $\ell \geq 1$ be fixed integers. We can in polynomial time decide if $\kappa_S^c(D) \geq \ell$ for any symmetric digraph D with $S \subseteq V(D)$, where $|S| = k$.*

Proof. Let $k \geq 2$ and $\ell \geq 1$ be fixed integers, and let D be a symmetric digraph. Let $S \subseteq V(D)$ with $|S| = k$, and let $\{A_i \mid 0 \leq i \leq \ell\}$ be a partition of $A[S]$, where $A[S]$ denotes the arc set in $D[S]$. We need to prove the following claim.

Claim. We can in time $O(n^3 k^{\ell})$ decide if there exists a set of ℓ internally disjoint S -cycles, say $\{C_i \mid i \in [\ell]\}$, such that $A(C_i) \cap A[S] = A_i$ for each $i \in [\ell]$ (note that A_0 denotes the set of arcs in $A[S]$ not used in any of the cycles).

Proof of the claim. Let C be any S -cycle in D . We define the *skeleton* of C as the cycle we obtain from C by contracting all vertices in $V(C) \setminus S$. Let C^S be a skeleton of an S -cycle in D . Note that $V(C^S) = S$ and therefore there are at most k^k different skeletons of S -cycles in D .

Our algorithm will try all possible ℓ -tuples, $C^S = (C_1^S, C_2^S, \dots, C_\ell^S)$, of skeletons of S -cycles and determine if there is a set of ℓ internally disjoint S -cycles, say $\{C_i \mid i \in [\ell]\}$, such that $A(C_i) \cap A[S] = A_i$ and C_i^S is the skeleton of C_i for each $i \in [\ell]$. If such a set of cycles exist for any C^S , then we return this solution, and if no such set of cycles exist for any C^S , then we return that no solution exists. We will prove that this algorithm gives the correct answer and compute its time complexity.

If our algorithm returns a solution, then clearly a solution exists. So now assume that a solution exists and let $\{C_i \mid i \in [\ell]\}$ be the desired set of internally disjoint S -cycles. When we consider $C^S = (C_1^S, C_2^S, \dots, C_\ell^S)$, where C_i^S is the skeleton of C_i , our algorithm will find a solution, so the algorithm always returns a solution if one exists.

Given such an ℓ -tuples, C^S , we need to determine if there is a set of ℓ internally disjoint S -cycles, say $\{C_i \mid i \in [\ell]\}$, such that $A(C_i) \cap A[S] = A_i$ and C_i^S is the skeleton of C_i for each $i \in [\ell]$. We first check whether $A_i \subseteq A(C_i^S)$. If this does not hold, then the desired cycles do not exist, so we assume that this holds in the following argument. For every arc $uv \notin A[S]$ that belongs to some skeleton C_i^S , we want to find a $u - v$ path in $D - A[S]$, such that no internal vertex on any path belongs to S or to a different path. This can be done in $O(n^3)$ time by Lemma 3.5. If such paths exist, then we obtain the desired S -cycles by substituting each uv by the corresponding $u - v$ path. Otherwise, the desired set of S -cycles does not exist.

Therefore, the algorithm works correctly and has complexity $O(n^3 k^{\ell})$ since the number of different ℓ -tuples, C^S , that we need to consider is bounded by the function $(k^k)^\ell = k^{k\ell}$. ■

By the claim, we can in time $O(n^3 k^{\ell})$ decide if there exists a set of ℓ internally disjoint S -cycles, say $\{C_i \mid i \in [\ell]\}$, such that $A(C_i) \cap A[S] = A_i$ for each $i \in [\ell]$. We will now use the algorithm of the claim for all possible partitions $\mathcal{A} = \{A_i \mid 0 \leq i \leq \ell\}$. If we find the desired set of ℓ internally disjoint S -cycles for any such a partition, then we return " $\kappa_S^c(D) \geq \ell$ ". Otherwise, we return " $\kappa_S^c(D) < \ell$ ". Note that if $\kappa_S^c(D) \geq \ell$, then we will correctly determine that $\kappa_S^c(D) \geq \ell$, when we consider the correct partition $\mathcal{A}$, which proves that the above algorithms will always return the correct answer.

Hence, we deduce that the algorithm works correctly and has complexity $O(n^3 k^{\ell} (\ell + 1)^{k^2/2})$ which is a polynomial in n , since the number of partitions $\mathcal{A}$ of $A(D[S])$ is bounded by $(\ell + 1)^{|A(D[S])|} \leq (\ell + 1)^{k^2/2}$, and both of k and ℓ are fixed. □

By Theorem 3.6, we directly have the following result.

Corollary 3.7. *We can in polynomial time decide if a given symmetric digraph D is k -cyclic, for a fixed integer $k \geq 2$.*

4. Discussions

The perfect forest theorem in undirected graphs shows that every connected graph of even order contains a 0-perfect forest [2,6,14]. However, there is no similar result on i -perfect out-forest for digraphs ($i \in \{0, 1\}$), that is, not every strong connected digraph of even order has a perfect out-forest, and not every strong connected digraph of odd order has a 1-perfect out-forest, as any symmetric digraph D contains neither perfect out-forest nor 1-perfect out-forest. Hence, it would be interesting to give some nice sufficient conditions or characterizations to guarantee the existence of an i -perfect out-forest in digraphs ($i \in \{0, 1\}$).

It would be interesting to consider the complexity for deciding whether $\lambda_S^c(D) \geq \ell$ on digraph classes, such as Eulerian digraphs. Note that the construction in Theorem 3.3 may not make D' Eulerian even when H is Eulerian, since

$\deg_D^-(u^-) (= \deg_D^-(u))$ maybe larger than $\deg_D^+(u^-) (= 1)$. Therefore, we need to find other approach to prove the NP-completeness for $\lambda_S^c(D)$ for an Eulerian digraph D . Certainly, we cannot exclude the possibility that deciding whether $\lambda_S^c(D) \geq \ell$ restricted to Eulerian digraphs is polynomial-time solvable.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We are thankful to Professor Anders Yeo for discussions on the perfect forest problem. Yuefang Sun was supported by National Natural Science Foundation of China under Grant No. 12371352, Zhejiang Provincial Natural Science Foundation of China under Grant No. LY23A010011 and Yongjiang Talent Introduction Programme of Ningbo under Grant No. 2021B-011-G. Zemin Jin was supported by Natural Science Foundation of China under Grant No. 12471318.

Data availability

No data was used for the research described in the article.

References

- [1] J. Bang-Jensen, G. Gutin, *Digraphs: Theory, Algorithms and Applications*, second ed., Springer, London, 2009.
- [2] Y. Caro, J. Lauri, C. Zarb, Two short proofs of the perfect forest theorem, *Theory Appl. Graphs* 4 (1) (2017) article 4.
- [3] J. Cheriyan, M. Salavatipour, Hardness and approximation results for packing steiner trees, *Algorithmica* 45 (2006) 21–43.
- [4] M. Grötschel, A. Martin, R. Weismantel, Packing steiner trees: separation algorithms, *SIAM J. Discrete Math.* 9 (1996) 233–257.
- [5] M. Grötschel, A. Martin, R. Weismantel, The Steiner tree packing problem in VLSI design, *Math. Program.* 78 (1997) 265–281.
- [6] G. Gutin, Note on perfect forests, *J. Graph Theory* 82 (3) (2016) 233–235.
- [7] G. Gutin, A. Yeo, Note on perfect forests in digraphs, *J. Graph Theory* 85 (2017) 372–377.
- [8] G. Gutin, A. Yeo, Perfect forests in graphs and their extensions, *J. Graph Theory* 101 (3) (2022) 572–592.
- [9] D. Kühn, D. Osthus, Linkedness and ordered cycles in digraphs, *Combin. Probab. Comput.* 17 (2008) 689–709.
- [10] X. Li, Y. Mao, *Generalized Connectivity of Graphs*, Springer, Switzerland, 2016.
- [11] N. Robertson, P. Seymour, Graph minors XIII: The disjoint paths problem, *J. Combin. Theory Ser. B* 63 (1) (1995) 65–110.
- [12] T.J. Schaefer, The complexity of satisfiability problems, in: J. Richard, Walter A. Lipton, Walter J. Burkhard, Emily P. Savitch, Friedman, Alfred V. Aho (Eds.), *Proceedings of the 10th Annual ACM Symposium on Theory of Computing*, (1978) 1–3, San Diego, California, USA, ACM, 1978, pp. 216–226.
- [13] A. Schrijver, *Combinatorial Optimization: Polyhedra and Efficiency*, Springer, 2003.
- [14] A. Scott, On induced subgraphs with all degrees odd, *Graphs Combin.* 17 (3) (2001) 539–553.
- [15] R. Sharan, A. Wigderson, A new NC algorithm for perfect matching in bipartite cubic graphs, in: *Fourth Israel Symposium on Theory of Computing and Systems*, ISTCS 1996, Jerusalem, Israel, (1996) 10–12, Proceedings, IEEE Computer Society, 1996, pp. 202–207.
- [16] N. Sherwani, *Algorithms for VLSI Physical Design Automation*, third ed., Kluwer Acad. Pub., London, 1999.
- [17] K. Steiglitz, P. Weiner, D. Kleitman, The Design of Minimum-Cost Survivable Networks, *IEEE Trans. Circuit Theory* 16 (4) (1969) 455–460.
- [18] Y. Sun, G. Gutin, Strong subgraph connectivity of digraphs, *Graphs Combin.* 37 (3) (2021) 951–970.
- [19] Y. Sun, G. Gutin, A. Yeo, X. Zhang, Strong subgraph k -connectivity, *J. Graph Theory* 92 (1) (2019) 5–18.
- [20] Y. Sun, G. Gutin, X. Zhang, Packing strong subgraph in digraphs, *Discrete Optim.* 46 (2022) Article 100745.
- [21] Y. Sun, A. Yeo, Directed Steiner tree packing and directed tree connectivity, *J. Graph Theory* 102 (1) (2023) 86–106.